

How asset tracking saves cost

Complexity in supply chain comes from the volume of items that is consistently in transit. Manual monitoring on a large scale leads to multiple challenges and, in turn, major financial losses. Yugank Goyal, chief executive officer, Riyanix, says, "Tracking helps in reducing risks, increasing predictability and forecasting, creating smarter routes, offers real-time control of the movement, increases efficiency, reduces delays and, consequently, reduces cost." Using these technologies, businesses can cut down operational expenses by at least 15 per cent.

Functional applications of asset tracking technology in supply chains is multifold. Goyal adds, "Benefits of inventory tracking for private businesses as well as the public sector are immense. Consider the huge losses Food Corp. of India incurs due to loss and theft of food grains. Think about the rampant theft in coal transport. Railways is another industry that can benefit hugely from tracking solutions. Vehicle tracking and fleet management is another recognised application. It could save additional costs by reducing the fuel consumption by providing optimal speed to improve mileage."

Setup and payback

A tracking solution setup depends on the scale of accuracy required (within a few centimetres), type of location where the setup is to be done and geographical range across which tracking needs to be carried out. As part of the setup, basic tracking modules and sensors need to be installed wherever tracking data is required, be it inside the equipment, around the premises or inside vehicles. A gateway device is configured to channel the tracking data to the Cloud.

Goyal talks about configuring their tracking solution, "Our solution gateway can be configured over various connectivity technologies including GPS, Bluetooth Low Energy (BLE) and LoRa. This allows many devices to communicate to the Cloud parallelly. On the mobile network, it directly communicates with the GSM network and transmits all data."

While the cost of investment varies



Asset tracking solutions can be highly beneficial for businesses (Credit: Wikipedia)

widely with the scale of the application, paybacks are quicker on bigger applications. Goyal adds, "For large-scale investments, tracking almost always makes economic sense. The larger the concern and costlier the losses in logistics, the quicker the return on investment when the solution is finally applied. Even for a mid-scale company, investment in tracking solutions is economically viable, with typical returns on investment (ROI) reached within one year."

Challenges in implementation

Adoption of tracking solutions in businesses is on a rise. A main challenge lies in awareness. Goyal explains, "The workforce needs to accept the technology. For instance, drivers assigned to company trucks need to adopt to the mobile devices and the electronically-logged devices inside their vehicles. Proper training and awareness needs to be given for efficient use of such systems."

The government can also play a crucial role in fast-pacing its adoption, starting from implementation in public infrastructure. Goyal says, "In my opinion, implementing CAN Bus protocol must be made mandatory in India for all public fleet of buses and trucks, especially in logistics. It will help build Internet of Things (IoT)

architecture in all these vehicles for remote communication, monitoring and tracking."

Success stories

Over the recent years, there have been a number of successful deployments of such solutions. For example, Zomato streamlined their services and solved logistic challenges using connected technology. The company wanted to address the inconsistent visibility of their vehicles during delivery and gather accurate information regarding delivery time and location. They took up proprietary GPS trackers, which were installed in their vehicles. This enabled Zomato to achieve 24x7 real-time visibility of their vehicles and delivery status.

Additionally, customers also receive real-time location status of the delivery. This has greatly benefited Zomato by cutting down error-prone or delayed deliveries and substantially reducing expenses. Quick service and transparency to customers have greatly improved its customer base and brand satisfaction.

Goyal gives another example of a US-based fleet management firm that installed Riyanix's asset tracking solution for vehicle management. He explains, "This consists of our VEANGEL gateways embedded inside